



KARIM AIT HAMMOU

Senior Engineer
Python / Django

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LINKS

- in linkedin.com/in/karimaithammou
- github.com/HumanZ19
- humanz19.github.io

PYTHON BACKEND

Python

Django

Django Ninja

Wagtail

FastAPI

REST APIs

PostgreSQL

Redis

Celery

Elasticsearch

APPLIED AI & LLM

Production RAG

Self-hosted LLM

pgvector

Ollama

OpenAI API

LangGraph

MCP

HuggingFace

Embeddings

ChromaDB

DELIVERY

Docker

Linux

CI/CD

AWS

Git

Testing

Documentation

LANGUAGES

French

Bilingual

Arabic

Native

English

Professional

Amazigh

Native

Karim AIT HAMMOU

Senior Python / Django Engineer · AI & LLM

AVAILABLE IMMEDIATELY · REMOTE OR RELOCATION

20 years building software, 5 of them on production LLM systems. Django backends serving as the application layer for language models: RAG, pgvector, self-hosted LLMs. Available immediately.

PROFILE

I build Python and Django backends that serve as the **application layer for production LLM systems**. On **PhytochemHub**, a Django Ninja API sits on top of a five-stage pipeline built around a self-hosted model, with embeddings in PostgreSQL/pgvector and no cloud dependency on the critical path. On **Sci-X**, a full RAG pipeline manages its own embedding lifecycle: invalidation, scheduled re-indexing, drift monitoring. On **KMB Trader**, a LangGraph multi-agent system runs 24/7. On the web side, I ship **Wagtail to production**, including two sites launched in 2026. Twenty years in IT behind that, including a stint as Chief Information Officer: I know how to take a system from prototype to operations, document it and hand it over. Available immediately, for contract or permanent work.

CORE COMPETENCIES

Django in production

Django, Django Ninja, Wagtail, FastAPI. REST APIs, durable data models, PostgreSQL, Redis, async tasks. Sites and APIs that run, not prototypes.

LLMs applied to the backend

Embedding models into a real application: full RAG, embeddings and pgvector, self-hosted LLMs via Ollama, OpenAI and HuggingFace APIs. The model is a component of the system, not a demo.

Lifecycle and reliability

Embedding invalidation and re-indexing, drift monitoring, supervision, structured logging, alert thresholds, human escalation. What keeps a system alive after launch.

Wagtail and editorial content

Wagtail sites delivered to non-technical clients, finely editable content, multilingual, SEO. Migrations away from WordPress and Drupal.

Autonomous delivery

Docker, Linux, CI/CD, AWS, DNS and hosting management. Able to take an engagement end to end with no team around me, and document it for whoever comes next.

Business counterpart

Twenty years translating business needs into systems. Former CIO: I speak to developers and to executives alike, which shortens decision cycles.

20

YEARS
BUILDING
SOFTWARE

5

LLM SYSTEMS
IN PRODUCTION

50+

PROJECTS
DELIVERED

+150%

ORGANIC
TRAFFIC
(ATELIERS
D'ART)

ASAP

IMMEDIATE
AVAILABILITY

PROFESSIONAL EXPERIENCE

Freelance - Python backend & AI

Jan 2021 - Present

Senior Python / Django Engineer - International clients

Design and delivery of Django backends, several of which serve as the application layer for language models in production.

- **PhytochemHub** (2026): scientific extraction platform. **Django Ninja REST API** on top of a five-stage pipeline: GPU PDF parsing, extraction by a self-hosted LLM (Ollama), 768-dimension embeddings in **PostgreSQL / pgvector**, real-time dashboard. No cloud dependency on the critical path, every data point traceable to its source.
- **Leaders On Purpose** (2026): two **Wagtail** sites shipped to production, migrated off WordPress, infrastructure and DNS zone administered, content editable atom by atom by a non-technical client.
- **Sci-X and Green Hydrogen**: complete RAG pipeline in Django. Adaptive semantic chunking, OpenAI and HuggingFace embeddings, pgvector and ChromaDB storage, full lifecycle management (invalidation, re-indexing, drift monitoring).
- **KMB Trader**: Python multi-agent system orchestrated with LangGraph, running 24/7 with real-time state management, alert thresholds and human escalation.
- **Ateliers d'Art de France**: Django and Wagtail architecture for 6,000+ professionals, +150% organic traffic.
- **WorldWide Team**: real-time geolocation platform in Django, event-driven architecture on WebSocket and Redis.

Python

Django

Django Ninja

Wagtail

PostgreSQL

pgvector

Ollama

LangGraph

Redis

Docker

Linux

Eduka Software

Oct 2025 - Feb 2026

Technical & Functional Project Manager - International SaaS ERP (10+ countries)

Integration architecture on a SaaS ERP in a regulated financial environment.

- Designed abstraction layers isolating business logic from per-jurisdiction implementations.
- Third-party API integrations with **transactional idempotency** on flows where a duplicate submission is a regulatory offence: e-invoicing (Spain, Brazil, Saudi Arabia, Morocco) and payments (Payway, TouchPay).
- Multi-site Proxmox and Teleport infrastructure across three countries, Zero Trust principles, critical incident resolved in 25 minutes.

PHP

MySQL

REST APIs

Proxmox

Teleport

Linux

HEM Business School (LCI Education Group)

Jan 2017 - Dec 2020

Chief Information Officer - 6 campuses

IS leadership of a multi-site institution. Relevant here for one precise reason: I know what a badly handed-over system costs.

- Critical ERP migration with zero downtime, 1 M MAD budget defended before a non-technical board.
- Managed 21 people, trained over 300 users.

Konosys ERP

Linux

Microsoft 365

HEM Business School

Jan 2010 - Dec 2017

Senior Webmaster then Deputy CIO

- Multi-site institutional web ecosystem (Drupal, PHP), managed a team of 10.

Drupal

PHP

MySQL

Sunlight Systems SARL

2007 - 2010

Founder & Technical Manager - Agadir, Morocco

- Three web products shipped in full autonomy, including SoussAgri.com, 2nd INNOFEL Prize 2008.

FLAGSHIP WORK

DJANGO + LLM

PhytochemHub

Django Ninja API on top of a self-hosted LLM extraction pipeline. PostgreSQL and pgvector, 768d embeddings, real-time dashboard. Live at phytochemhub.com.

Django Ninja · Ollama · pgvector · Docker

WAGTAIL IN PRODUCTION

Leaders On Purpose

Two Wagtail sites shipped and maintained in production, migrated off WordPress, infrastructure and DNS included. Fully editable by a non-technical client.

Wagtail · Django · cPanel · SEO

FULL RAG

Sci-X and Green Hydrogen

RAG pipeline in Django: adaptive semantic chunking, OpenAI and HuggingFace embeddings, pgvector and ChromaDB, scheduled invalidation and re-indexing.

Django · OpenAI · pgvector · ChromaDB

PYTHON MULTI-AGENTS

KMB Trader

LangGraph-orchestrated multi-agent system, 24/7, real-time state management, alert thresholds and human escalation.

Python · LangGraph · Redis · Docker

DJANGO AT SCALE

Ateliers d'Art de France

Django and Wagtail architecture for a directory of 6,000+ professionals. +150% organic traffic after the rebuild.

Django · Wagtail · PostgreSQL · SEO

TOOLING

Wagtailer

Agent-driven Wagtail industrialisation framework: thirteen-phase pipeline, executable guardrails, client-agnostic project scaffold.

Wagtail · Python · Agents · Git hooks

WHAT I BRING

The Django and LLM overlap

Most Django developers have never put a model in production, and most AI profiles cannot keep an application running. I do both, and it is verifiable on live code.

Sovereignty by default

Self-hosted LLM pipelines, local GPU, no data sent to any third party. Decisive for healthcare, legal, finance and the public sector.

Full autonomy

I take an engagement from need to launch: design, code, infrastructure, documentation, handover. With no team to assemble around me.

Available immediately

No notice period to serve. I can start within the week, remote or with relocation.

EDUCATION

ESI (IHEES)

Computer Management & Network Development
2002 - 2006

Abdellah Ben Yacine High School

Baccalaureate Mathematical Sciences
1999 - 2002